MAKING SENSE OF GREEN LUMBER

What does FSC certification mean for the future of sustainable lumber?

by ARMSTEAD FELAND V

In recent months, I have been building decks in Northern California out of a Brazilian hardwood known as ipe. The client on one deck project was pleased with the tropical hardwood because of its beauty, and the project manager was happy that the use of this extremely dense hardwood increased the amount of time needed to finish the job (which translates to more money). Ipe is 3 times harder than oak, and is now used as a replacement for redwood, which most homeowners can no longer afford. Known as the poor man's teak, ipe is enjoying a new popularity in California.

However, I was alarmed to learn that the ipe we were using had been shipped a considerable distance and might not have been harvested in a sustainable manner. But on tracking down the supplier, I discovered that our ipe was indeed Forest Stewardship Council (FSC) certified. The FSC, an international organization that promotes responsible management of the world's forests, relies on a dozen third-party certifiers worldwide, two of which—Smartwood and Scientific Certification Systems (SCS)—are located in the United States. Strict sets of standards ensure that FSC-certified wood is ethically and environmentally sustainable.

Once I informed the clients and contractor that our ipe was sustainable, everyone breathed a sigh of relief, and they all felt good about themselves once more. But the fact remains that it takes a significant amount of energy to transport these timbers from the rain forests of Brazil to the hills of Berkeley, California.

CHARLY RAY

Extremely rot resistant and native to the United States, FSC-certified black locust wood is a good choice for exterior applications.

Why FSC?

Once a material has been certified as sustainably harvested, does that automatically make the material ecologically viable? What exactly does certification of wood products mean? Does it simply allow consumers to believe that they are making the best choice possible for their green home without ever considering the fact that they have purchased a nonlocal product, the harvesting of which may or may not have serious environmental and ethical implications?

In fact, the quality and meaning of the green certification depends on the program that certifies the lumber. Several programs in the United States offer different levels of certification for lumber. One of these is the Sustainable Forestry Initiative (SFI), sponsored by the American Forest & Paper Association (AFPM). The members of this trade group, who comprise 90% of the industrial timberland in the United States, are required to participate in the SFI. The SFI sets broad standards and it requires members of AFPM to file annual reports showing how they are complying with these standards. The SFI has provided additional training for loggers and has given foresters a greater say in how companies operate. However, the SFI allows clear cuts of up to 120 acres, while the FSC limits clear-cutting to 40 acres in most areas of the United States and even less in some parts of the country. The SFI approves of the planting of genetically modified trees, while the FSC favors natural selection and allows only traditional seed-breeding techniques. In addition, the SFI only requires companies to obey the laws governing the use of pesticides, while the FSC requires companies to reduce their use of pesticides. And while the SFI puts the landowners in charge, the FSC gives environmentalists and community groups a say in how its certified forests are run. Furthermore, the SFI fails to include many elements that environmentalists consider necessary for credible certification and product labeling. SFI-certified companies, such as Weyerhaeuser and Pacific Lumber, contribute to the destruction of forests and wildlife habitat by using such practices as logging or buying wood from old-growth or endangered forests; replacing natural forests with ecologically barren pine plantations and sprawl; and using large clear-cuts, toxic herbicide spraying, and other harmful logging practices.

While any set of standards that encourages landowners to implement more sustainable solutions to forestry management is an improvement on typical large-scale forestry management practices, clearly not all programs are created equal. So far, FSC standards have proven themselves to be the most reliable and environmentally sensitive. The lumber that carries the FSC decal is guaranteed to come from a well-managed forest—one that hasn't been severely clear cut, genetically modified, or liberally sprayed with herbicides. "SFI brings some important contributions, but it's not the same," says Robert Hrubes, senior vice president of SCS. "The metaphor that's been used to describe SFI is that it's a rising tide that floats all ships. I think that's a very apt description. Everyone that's a dues-paying member can get certified. Nonetheless, it does rise the tide a little bit, and everyone in the trade association is doing a little better forestry practices as a result."

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The FSC has certified over 10 million acres of managed natural forests and plantations worldwide.

Getting Certified

The FSC relies on a certification process that is an independent, scientific, third-party verification of guaranteed sustainable forestry, with chain-of-custody tracking on the resulting wood products. This process of scientifically quantifying and defining forestry management systems helps to ensure that clients are connected to the source of their wood products. Thus far, over 10 million acres of managed natural forests and plantations have been certified worldwide, and over 635 million ft of certified wood are traded annually.

The certification is based on internationally applicable principles and criteria that have been established by the FSC. Certification can come in the form of forest management certification or chain-of-custody certification. Forest management certification means that the forest products company is responsible for understanding and carrying out forest management practices in a manner that meets or exceeds the standards of the certification system, while chain-of-custody certification assures consumers and forest products companies that the wood they buy comes from certified forests. Chain-of-custody certification verifies the flow of forest products certified by the FSC through the supply chain, from the forest to the point of sale. Chain-of-custody certification is available for sawmills, secondary manufacturers, broker/distributors, wholesalers, retailers, printers, paper merchants, and other points in the forest products supply chain.

The principles and criteria for certification fall into three categories: including sustainable harvest, ecosystem health, and social and economic considerations. Certification was established to help all participating parties to avoid debate over management, harvesting, and conservation of the world's forests. Large box stores such as The Home Depot and Lowe's, are demanding more certified products from their suppliers.

Market viability is an important consideration for small operators who want to gain FSC certification. Liza Murphy, the forest products marketing associate at SmartWood, notes that while forest management certification may be out of many small operators' reach financially, chain-of-custody certification is a viable alternative. "Chain of custody is best thought of as inventory management," says Murphy. "It doesn't speak to the production process of the thing that the tree is turned into; it speaks to the credibility of the claim that the wood is coming from a well-managed forest. So a small producer is typically a potential chain-of-custody holder. The fee for chain of custody is divided into the flat fee portion and then the fee that differs depending on the revenue of the company. For a commercial provider, the fee ranges from $2,500 to a fairly unusual high fee of $5,000."

For small providers who are barely scraping by, this extra cost is often prohib-

Cooperating for Certification

On the south shore of Lake Superior, in northwest Wisconsin, the for-profit, member-owned Living Forest Cooperative (LFC) is helping to strengthen its community by practicing FSC-certified sustainable forestry, and by developing markets for sustainable forest products. Incorporated in 2000, the LFC has 127 members, most of whom are private landowners. The LFC manages 13,000 acres of forest, all of which are certified by the FSC; currently 4,000 acres are in active management.

While conventional approaches to forest management tend to favor short-term economic gain, the LFC focuses on the long-term ecological health and profitability of the forests it manages. By offering services such as forest stewardship plans, tree planting, ecosystem restoration, fire management, nuisance plant control, and a tool and supply library, the LFC provides a comprehensive sustainable management plan. The LSC takes into consideration all aspects of the forest it manages—not just timber production. Before any timber is harvested, the LSC generates a forest stewardship plan. This plan includes a detailed analysis of the timber resource; it addresses such issues as biodiversity, ecosystem health, and the aesthetic beauty of the landowner's forest. It explains how the landowner should care for the land, and it analyzes the land use and income potential of the property.

The timber and wood product market tends to be a "cut throat hard industry to get into," says Charly Ray, LFC's general manager. Since most individual landowners lack the time, experience, or money to manage their forest, let alone try to process and market their own wood products, the LFC combines its members' timber sales. This allows it to negotiate a higher stumpage price, which enables individual landowners to get the best possible price for their product. By collectively pooling members' timber, the LFC can also get better rates for milling, trucking, and so forth. This same cooperative effort goes into the LFC's marketing strategy, which enables members to get a higher price for raw and processed wood products than they could get if they tried to sell those products on their own.

LFC

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FSC certification means that the certified forest is managed according to strict environmental, social, and economic standards. The certification would cost an individual landowner $5,000 a year, but, because the LFC is a cooperative, members pay only $50 a year to ensure that their forests are certified FSC. As an added incentive, Ray says that "FSC certification provides an angle" in the timber industry. Recently Time Warner, Incorporated, told its paper suppliers that, starting this year, it would buy only sustainably certified paper. Since Wisconsin is ranked as the top paper-producing state in the nation, this push from Time Warner has put pressure on the pulpwood industry to obtain certification for its forests. Most of the wood harvested by the LFC is sold as pulpwood, and in the past five years they have harvested 15,000 pulp cords. Even though most of the certification in Wisconsin that has been prompted by Time Warner's ultimatum is through SFI certification, Ray still believes the ultimatum will help make FSC-certified pulpwood more marketable. Ray also foresees that in the future "private landholders might not be able to sell their wood without certification," if similar market trends are set in motion elsewhere.

The LFC has also tried its hand at processing and marketing its own timber products, such as flooring, paneling, and siding. But Ray says that it is hard to keep a steady inventory of material when you have only "just so much" of one product. The especially hard sell for local contractors and homeowners is that most local lumber yards can sell comparable products at a 20% markdown from the LFC's price.

In the years to come, the LFC hopes to grow from managing 13,000 acres of forested land to managing 20,000 acres. With an increase in consumer awareness and—the LFC hopes—more members, it may market certified processed timber products as well. For now, the LFC's focus is on developing a sound business model for its forestry management services. When a board member of the LFC asked Ray how far the cooperative could grow, Ray spoke of forest cooperatives in Sweden that not only manage the forest but own the mills. "There is huge potential," he says.

To learn more about the Living Forest Cooperative, go to www.livingforestcoop.com.

itive. Sometimes small producers must look at the big picture to decide whether or not they can afford to gain certification (see "Cooperating for Certification").

"I hear that [FSC certification is too expensive for small landowners] all the time. Frankly the truth is something different than that widely held misconception," says Robert Hrubes. "The FSC has had mechanisms in place for many years designed expressly to reduce costs for small landowners engaging in the FSC certification process. The FSC's basic policy that it adopted back in the mid-'90s is that the cost should not be a barrier to participation. The FSC has group certification, where aggregations of small landowners can be certified at one time, thereby reducing the costs to landowners. More recently, the FSC developed the Small and Low Intensity Forestry Management (SLIFM) program. This is a set of streamlined auditing protocols designed to reduce the cost—both the initial up-front cost and the ongoing maintenance costs associated with staying certified. For instance, for a small landowner the audit intensity is reduced up front, the report doesn't have to be peer reviewed, the size of the audit team can be reduced, and perhaps most substantively, small SLIFM operations don't have to undergo annual audits. The audits can take place every other year."

When I ran a timber frame company in northern Wisconsin, I ordered all of my timber material directly through local suppliers. Some of the suppliers were more sustainable than others in their harvesting practices. One supplier logged with horses, while another used a Skidster. But both loved the local forest and cared deeply about the regional ecosystem, about their community, and about finding ways that they could contribute to both. What I liked most about getting my lumber through these local suppliers was that by doing so I could contribute directly to my local economy base and help to control how my region was being affected, both economically and environmentally. I could easily have used a supplier in Montana who was FSC certified, but then I wouldn't have had the opportunity to influence my own region.

Liza Murphy agrees that local suppliers are the most sustainable source of lumber for home builders. "For things

CHARLY RAY

Many small operators may opt for chain-of-custody certification, since it is more affordable.

that you can source locally, [Smartwood] would agree and support that you should source them locally. We'd also go further and say for solid wood that you can source locally; you should work hard and see if you can get FSC-certified products, because when you've got almost 700 certificate holders [offering FSC-certified wood], you've got a lot of production capacity."

Murphy also says that using local tree species is a building strategy that simply makes economic sense. "If I'm building in the southeast part of the United States, then I'm typically building with southern yellow pine for standard stick framing. If I'm building in the northwest, I'm building with the local species there, whether it be hemlock fir or Douglas fir. That's a function of the marketplace doing what it does best, because it's less costly in that case. Once you know that you're going to use a locally produced material, look for [material that is] FSC certified."

Marketing locally may well be the smartest way for small landowners to make their FSC certification cost-effective. "Small landowners are at a disadvantage in market transactions with purchasers of their products," says Hrubes. "Size counts in markets, and small landowners historically, around the world, have a hard time deriving top dollar. This may well mean that one of the most effective ways to help small landowners realize the benefits of certification would be through some collective marketing of products at the regional level."

FSC and LEED for Homes

The new Leadership in Energy & Environmental Design (LEED-H) for Homes guidelines, whose yearlong pilot program was launched in the fall of 2005 by the U.S. Green Building Council (USGBC), builds on the incredible success of the existing LEED rating systems that focus on commercial buildings, and addresses the needs of the residential market. Builders can earn a maximum of 108 points on a home, with 90 points equaling LEED platinum status; 70 equaling gold; 50 equaling silver; and 30 points equaling generic LEED for Homes certification. Points can be earned in several categories for the use of FSC-certified wood. In fact, no other lumber certification program is recognized in LEED for Homes—the guidelines are based exclusively around FSC-certified lumber. "SFI put a lot of pressure on the USGBC to change the point credit systems so SFI-certified products get the same credit as FSC-certified products," Hrubes points out. "To LEED's credit, they've resisted that pressure and continue to maintain the recognition of FSC lumber. This fundamentally speaks to the fact that not all certifications are created equal. They are fundamentally different. FSC is the most rigorous and the most demanding and deserves special recognition in the marketplace."

One category where no points are given—but in which FSC certification is mandatory—is the use of tropical hardwoods. "The way that the LEED for Homes was originally written banned any tropical wood. I went back to them and said, 'Guys, I know what you're trying to do here, but let's pull the thread on this and dig a little deeper.' If we say that we're not going to use it, the use of tropical woods will still happen," says Smartwood's Murphy. "What we think is vital is [to] work with people and look at their practices around forests, and encourage and reward them for using the best practices. Tropical deforestation will not stop if LEED for Homes or anyone else bans tropical woods. It's becoming less and less appropriate to advocate for simply putting a fence around something, and it's becoming more and more imperative to figure out how we can engage with businesses that are interested and trying to do the right thing."

Under the category Materials and Resources, Credit #5, Environmentally Preferable Products, builders can earn a maximum of 4 points for the use of FSC lumber. Anywhere that lumber is used—for doors, windows, roofing, flooring, or walls—you earn a maximum of 0.5 points for each application. For instance, if you used all FSC-certified framing material, you'd earn 1/2 point; if you used all FSC-certified doors and windows, you'd earn an additional 1/2 point, and so on.

Granted, these numbers seem pretty small, given the potential 108 points that can be earned. However, there's a lot to consider in building a LEED-certified home. "LEED for Homes has many important goals that they've tried to balance with the creation of the standards. … Most residential building in the United States is done with stick-built framing. I think that it might be appropriate to make more points available for sustainable wood use," says Murphy.

At some floor area, a residential home turns into an unsustainable home, no matter what products you put into it, says Murphy. She notes that if two people are living in a 5,000 ft^2 home, the home cannot be considered green. "In one version, LEED for Homes came up with a baseline for an average home size. If you're at that size, you are required to get a certain number of points in the materials section. If you go smaller, then the mandatory number of materials points that you're required to get decreases. If you want to go larger, your mandatory number of materials points increases. In the pilot, they decoupled the house size from the materials section. By decoupling it from the materials section, I think that they weakened [the guidelines] a bit, because by tying it to the materials you're forcing people to look for other products that perhaps they wouldn't have unless they had to. So, by maxing out at four points for materials, and decoupling from a penalty for large homes, I think that we lost a really good opportunity for environmentally preferable materials. And not just wood, but windows, flooring, and the whole variety of materials. I'd like to see the guidelines recouple the penalty very closely with the materials."

The Future of Green-Certified Wood

I personally feel that the next step is to market more locally grown and produced lumber that is FSC certified. While it's reassuring that this is on the FSC's agenda as well, we're still a long way from achieving that goal, although important steps have been made. Globally, in the last ten years, FSC has hit 10% market penetration in around 60 countries.

Murphy says that this number needs to grow considerably—she'd like to see the rate of FSC-certified lumber double in the near future, and reach at least 90%. "For solid wood in particular I'd like to see residential builders—the Rylands, Sun Tecs, of the world—take on the opportunity through programs

Reclaimed Wood—Time-Tested, FSC Approved

There are a number of companies who specialize in FSC-certified reclaimed wood to provide for building needs. To meet certification, companies must show that their lumber comes from post-consumer recycled wood and wood fiber (such as, used pallets); trees from (sub)urban private and government properties (non-forested areas) that are dead, fallen, diseased, or a nuisance;trees from orchards that are unproductive and cut for replacement; wood recovered from deconstruction projects of buildings, railroads, and so on; and, wood recovered from demolition landfills.

Terra Mai

This company in northern California, reclaims wood that is currently destined for landfills, wood chippers, and burn piles around the world and use it to offset the demand for new lumber. Terra Mai is currently going through the application process to meet FSC certification through the SmartWood Rediscovered program.

For more information, visit www.terramai.com

CitiLogs

CitiLog, located in Pittstown, New Jersey, aims to prevent high quality urban trees from becoming mulch, firewood, or landfill waste. Instead, through salvaging/urban logging, horse logging, and FSC-certified wood for custom millwork, CitiLog is turning these trees into high quality lumber. CitiLog is FSC-certified and provides wood for historical restoration, cabinets/casework, molding/millwork, and flooring.

For more information, visit www.citilogs.com.

RAINFOREST ALLIANCE

FSC has attained 10% market penetration in over 60 countries.

like LEED for Homes to incorporate certified wood and preference certified-wood products."

Returning to our original question: Can tropical hardwoods qualify as green lumber? By now it should be obvious that the answer to this question is more complicated than simply calculating the number of miles that the wood has traveled from the forests of Brazil to the hills of Berkeley. "As with most things, the answer is, It depends," says Murphy. "Let's say someone wants to use tropical hardwoods for something like pilings. Tropical hardwoods have a series of characteristics that make them particularly appropriate for use in places where they're under a great deal of stress. The alternative has been to treat a piece of wood with chemicals to give it the same properties that that piece of tropical hardwood has. And, as we've seen from wood treated with arsenic on children's playgrounds, or marine pilings treated with chemicals, these compounds leech into the environment. And so what's the impact there versus an FSC-certified tropical hardwood?"

While this is true, it's also apparent that FSC certifiers have a stake in continuing the use of tropical hardwoods. Perhaps a more relevant question would be not whether FSC-certified tropical hardwood is better to use than chemically-treated wood, but whether there is a regional FSC-certified wood that would be even more sustainable than the other tropical hardwood or chemically treated alternatives.

And in fact, there is. Black locust, which is used primarily for fence posts, is just such a regional wood. Black locust is extremely rot resistant, due to flavonoids in the heartwood, and it can endure for over 100 years in the soil. Native to the United States, it grows indigenously on the lower Appalachian mountain slopes and in the Ozarks. Black locust wood has been widely used for fence posts, mine timbers, poles, and railroad ties in the past, but now it has found new uses as decking, sill plates, and various other exterior applications. Transplanted from its native range, it can be found in temperate forests around the country. Now viewed as an invasive species, due to its vigorous ability to reproduce by root suckering, it is still a good choice for outdoor applications. And you can easily find black locust that's FSC certified.

Given the new LEED for Homes requirements, and the growing demand for green lumber, FSC certification is becoming more desirable. Robert Hrubes points out that SCS recently FSC certified 4.8 million acres of state forest land in Minnesota and 3.9 million acres of state forest land in Michigan. "In the last year there's been a marked increase in the expansion of certified wood in the United States. I can't attribute it all to LEED, but I can tell you that it's an important ingredient in that trend."

When Liza Murphy is asked for her take on the future of green lumber, she says that she thinks it's bright. "I want to single out and recognize what the USGBC has done with the LEED program. There are a variety of other building programs out there—different states, different counties, have their own programs. Many of them follow the LEED guidelines, many of them tweak them for whatever their local need is; and so many of those [programs] preference FSC. But as a marketing influence organizing body, the support that USGBC showed for FSC-certified products has been a very important driver."

Murphy adds that some companies are starting to think about how they can talk to consumers about FSC certification. Getting certification seals on the wood is an important step, and even if only 1 out of 500 customers stops to examine the seal, that's still a beginning. And now that the process has officially begun, it seems inevitable that it will ramp up. The desirability of FSC-certified lumber is at an all-time high; even mainstream builders are beginning to recognize that consumers are interested in the product. And with recognition and desire comes market availability and viability. This means good news for everyone: FSC-certified forests and the people who operate them; the homeowners who buy the product; and every lumberyard and home building store that stocks the lumber. As Murphy notes, though it may not be advertised in neon letters, if you take a look at the SKU code on a piece of lumber, if the letters "FSC" are embedded somewhere in the code, you know that it's FSC-certified. Which is a good analogy for the current state of green lumber—it's not readily visible in the marketplace quite yet, but it's making a strong entry into the world of green materials. And with the new LEED for Homes guidelines, it looks like it's here to stay.

Armstead Feland V is a carpenter and timber-frame instructor at San Francisco Institute of Architecture (SFIA), where he is finishing a master's degree in ecological design.

FOR MORE INFORMATION:

To learn more about FSC certification, visit either www.scscertified.com or www.smartwood.org.